Introduction

Zhuoqi **Cheng**
zch@mmmi.sdu.dk
SDU Robotics

SDU

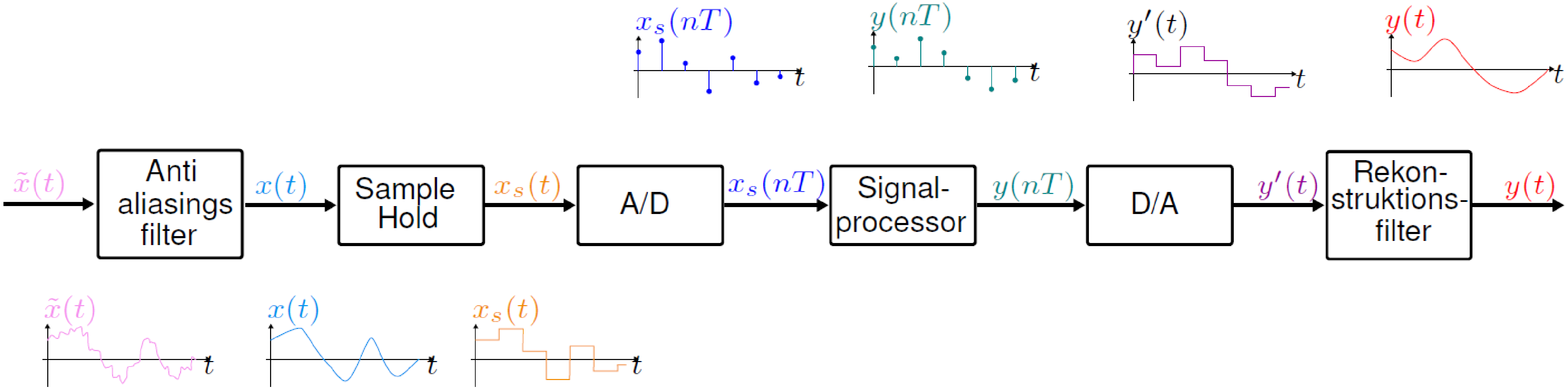
Topics to be covered in this course

→ Analog signal v.s. digital signal
→ Sampling and reconstruction
→ Aliasing
→ Fourier transform: time-frequency domain
→ Z transform
→ System analysis
→ Window functions
→ Filter and realization
→ Digital filter design(FIR and IIR)
→ Multi-rate sampling

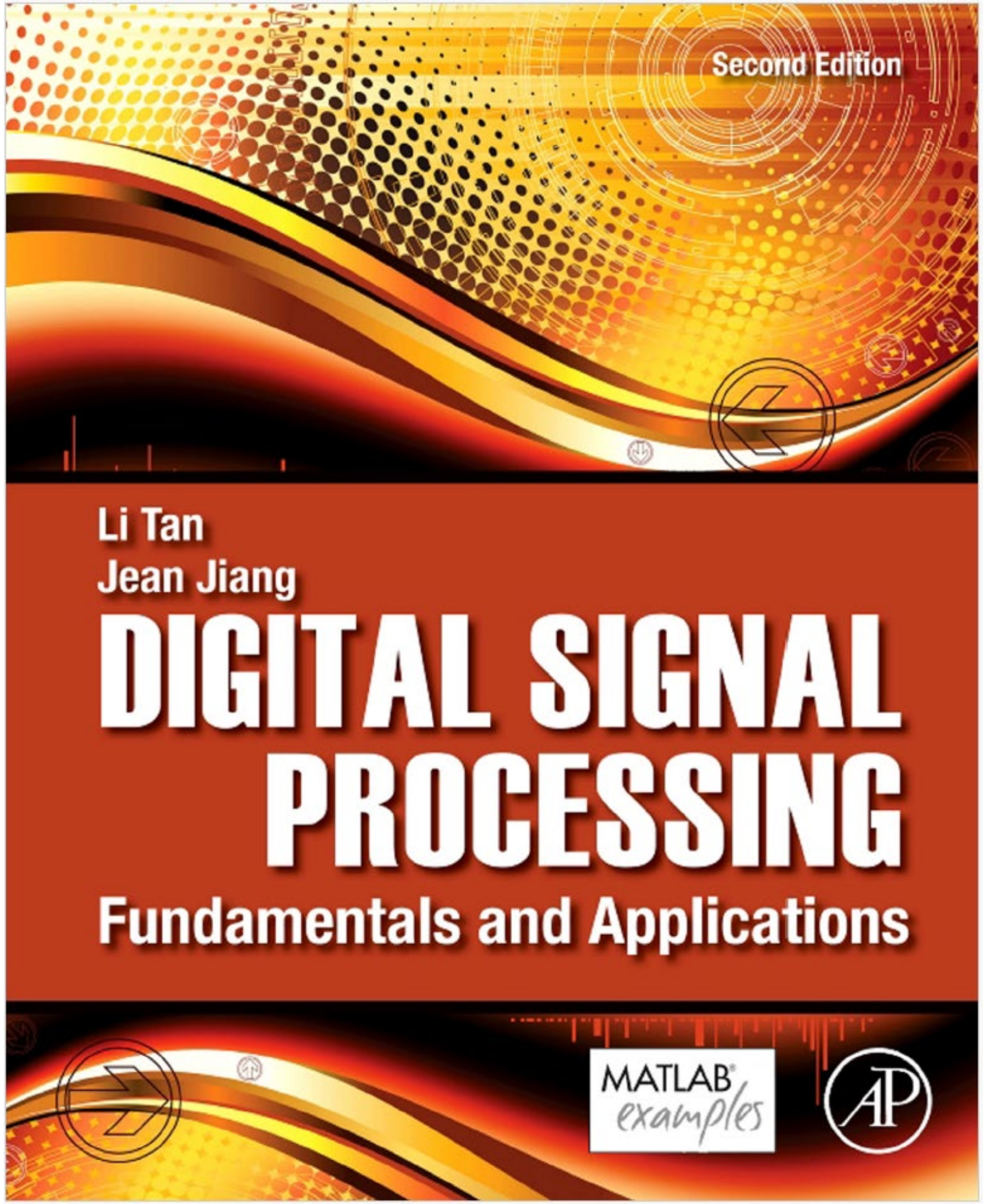
Lecture:
Zhuoqi Cheng
8.15 – 10.30 or 11.00

Exercise:
Nicolai Hedegaard Jensen
10.30 or 11.00 – 12.00

Digital signal processing workflow



Reference book



SDU

Topics NOT covered in this course

→ **All** Chapter 9. Hardware and Software for Digital Signal Processors
→ **Most of** Chapter 11. Waveform Quantization and Compression
→ **Part of** Chapter 12. Multirate Digital Signal Processing, Oversampling of Analog-to-Digital Conversion, and Undersampling of Bandpass Signals
→ **All** Chapter 13. Subband- and Wavelet-Based Coding
→ **All** Chapter 14. Image Processing Basics

Extra Topics not in the books

→ Matlab basic & DSP toolbox & Simulink tutorial
→ IIR – Matched z-transform
→ Detrend & Envelop operation

Reading materials will be provided!

About exam (written exam)

→ Multi-choice questions (only one answer is correct)
→ Calculations

digital signal v.s analog signal

Zhuoqi Cheng
zch@mmmi.sdu.dk
SDU Robotics

SDU

CHAPTER

1

Introduction to Digital Signal Processing

CHAPTER

3

Digital Signals and Systems

Reference book

Chapter 1: Introduction to Digital Signal Processing
(Page 1-13)

Chapter 3: Digital Signals and Systems
(Page 57-85)

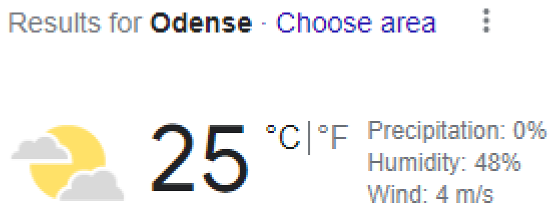
SDU

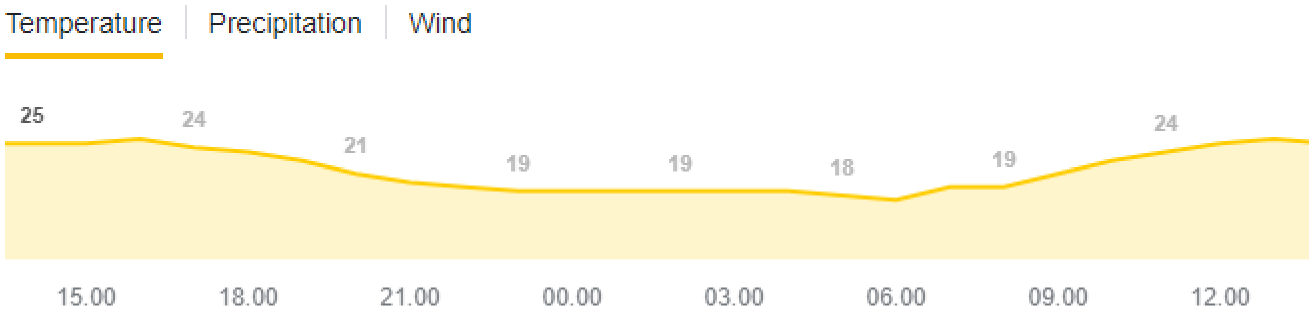
Signal

→ Analog v.s. Digital signal

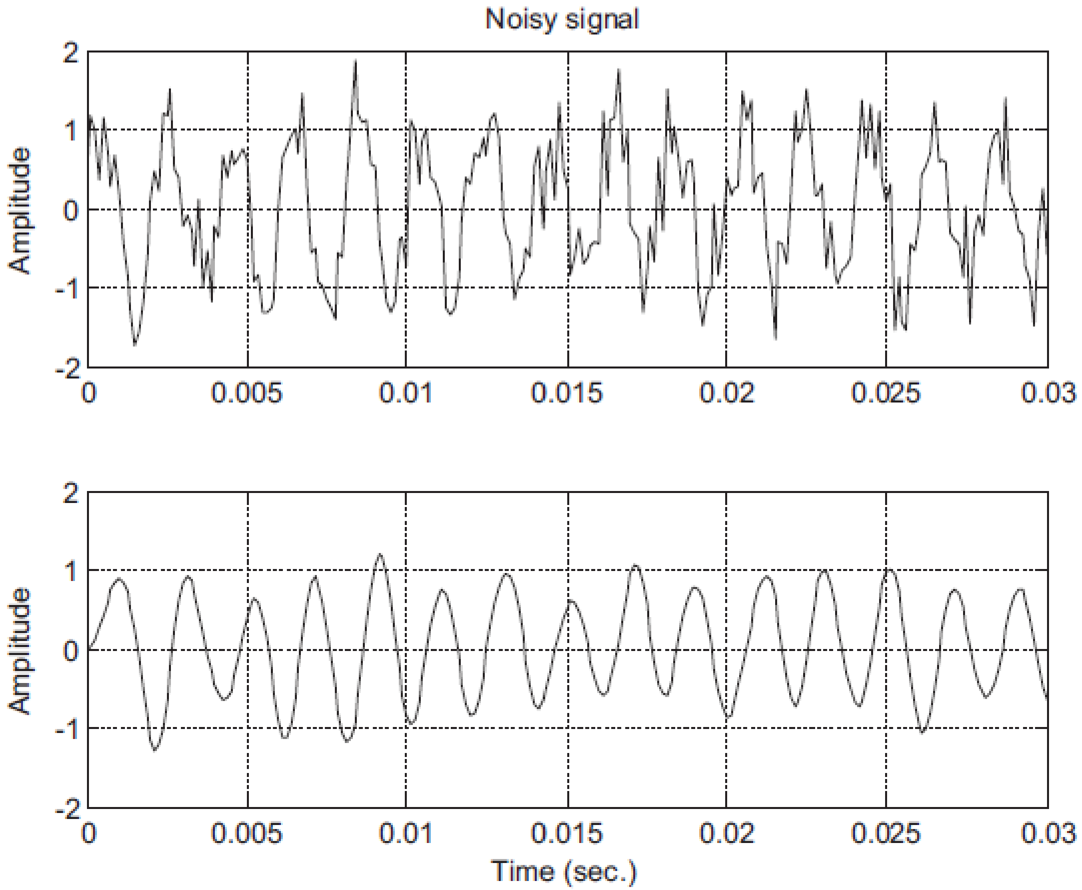
→ Continuous, discrete and series

→ Digital signal is a discrete time signal with quantization





Signal filtering



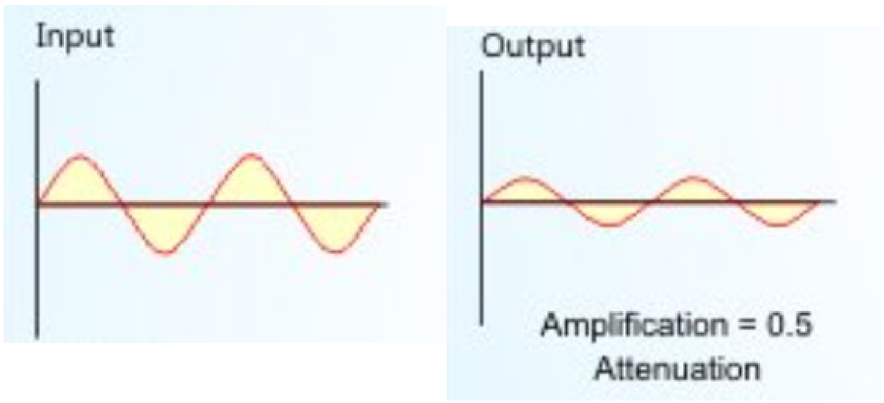
Analog signal & filter

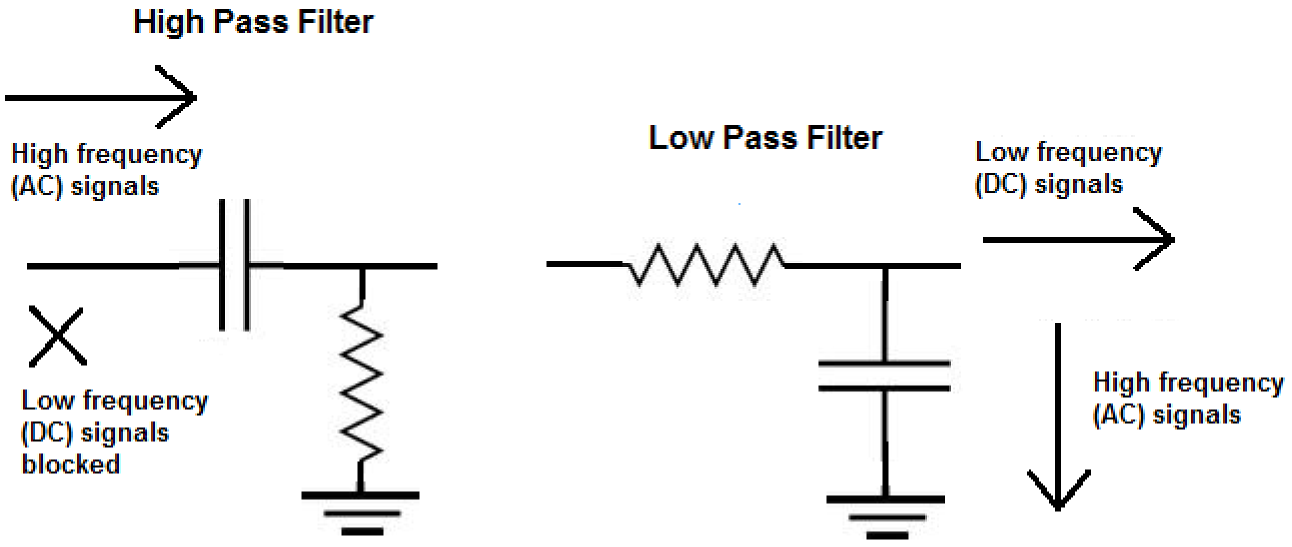
R-C circuit for <u>electric signal</u>

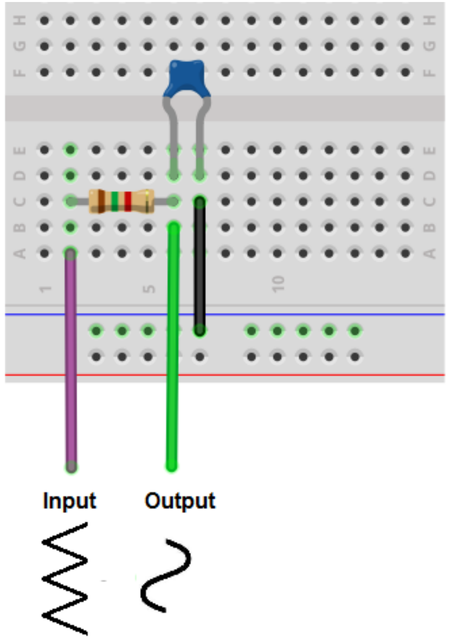
Cut-off frequency

$$f_c = \frac{1}{2\pi RC}$$

Cut-off frequency is defined as the frequency point where the output signal is attenuated to 70.7% of the input signal.

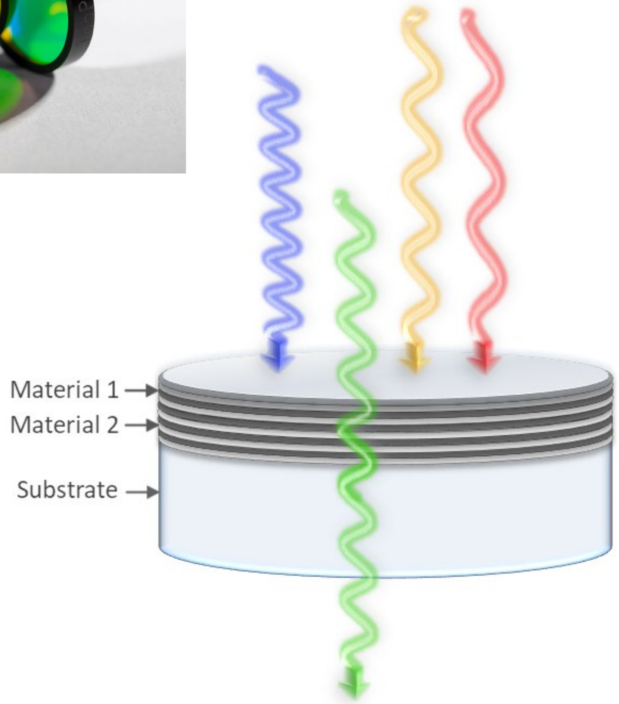


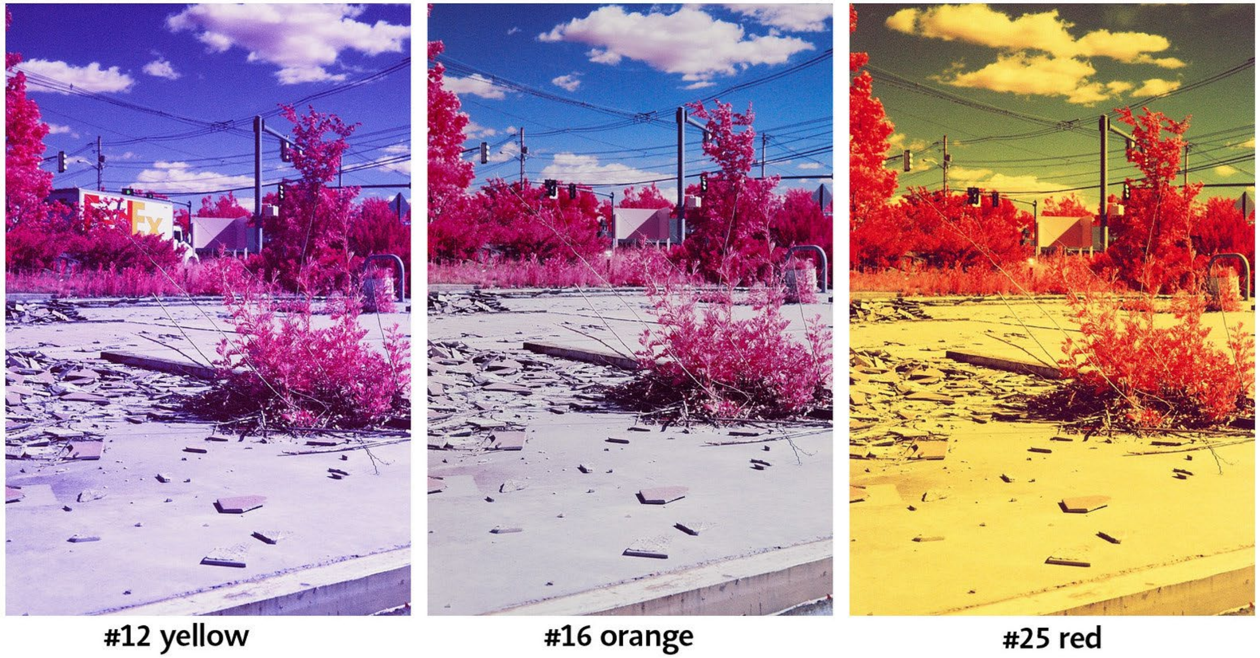




Analog signal & filter

Optical filter

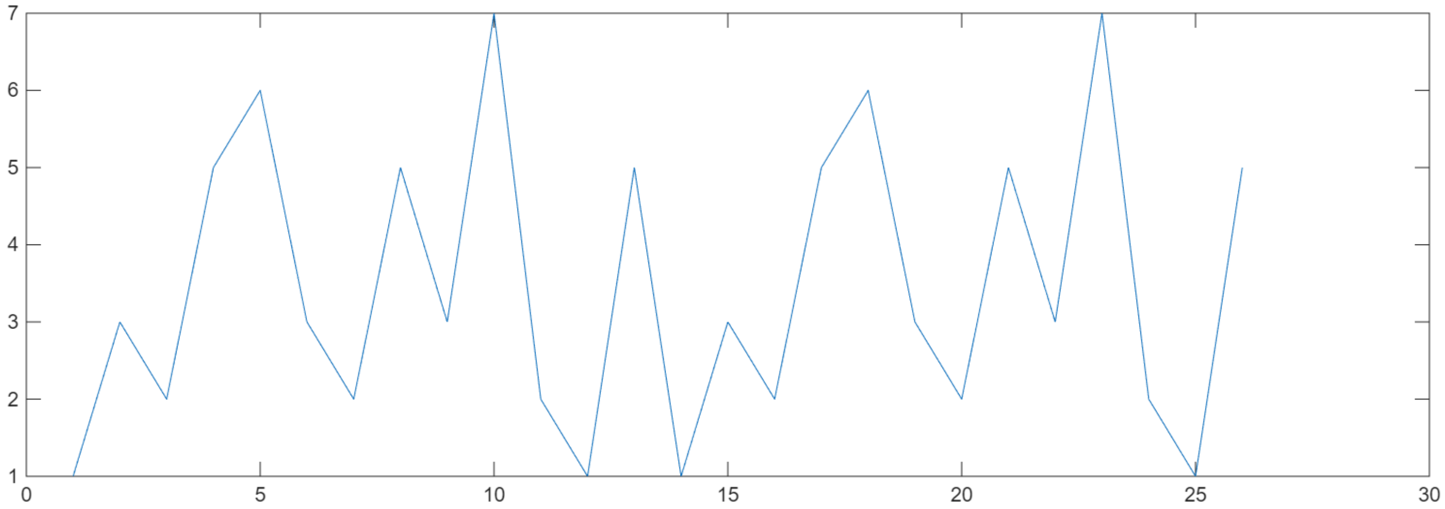




filter comparison for color infrared E-6 film

Digital signal

Digital signal = [1,3,2,5,6,3,2,5,3,7,2,1,5,1,3,2,5,6,3,2,5,3,7,2,1,5]



How can we do the signal filtering?

Discrete time signal – basic operation

$x(n)$ is the input digital signal

→Multiplier

$$y(n) = \alpha x(n)$$

→Adder

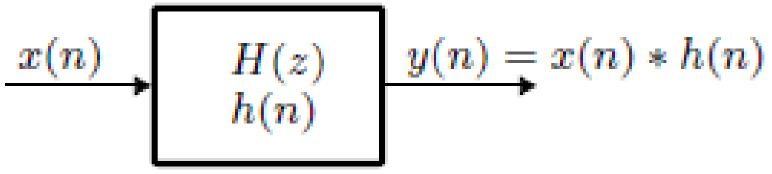
$$y(n) = x_1(n) + x_2(n)$$

→Delay

$$y(n) = x(n-1)$$

Discrete time convolution

Discrete time convolution can be used to calculate the output sequence for a time-discrete system using the input sequence $x(n)$ and the impulse response $h(n)$.

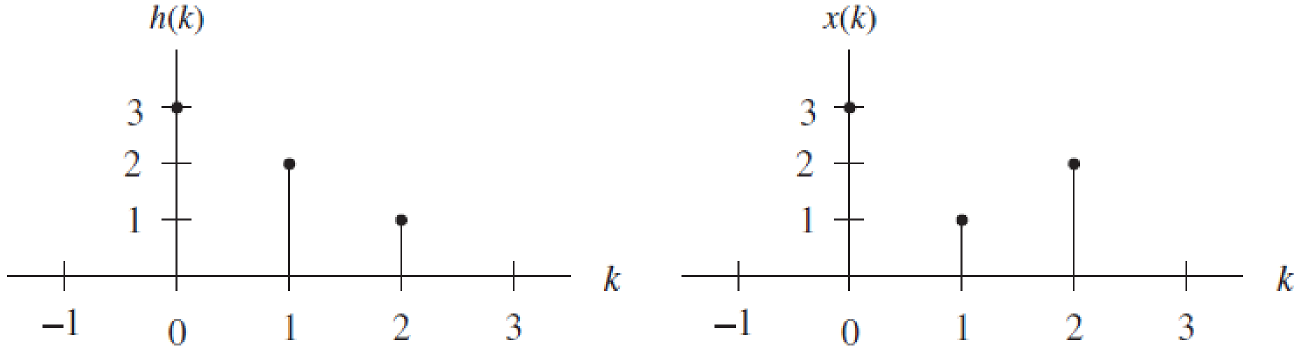


The output sequence $y(n)$ is calculated by

$$y(n) = h(n) * x(n) = \sum_{m=0}^{n} h(m)x(n-m)$$

Example

We consider a system with the following impulse response and input sequence.



Calculate the output sequence $y(n)$ using discrete time convolution

$$y(n) = h(n) * x(n)$$

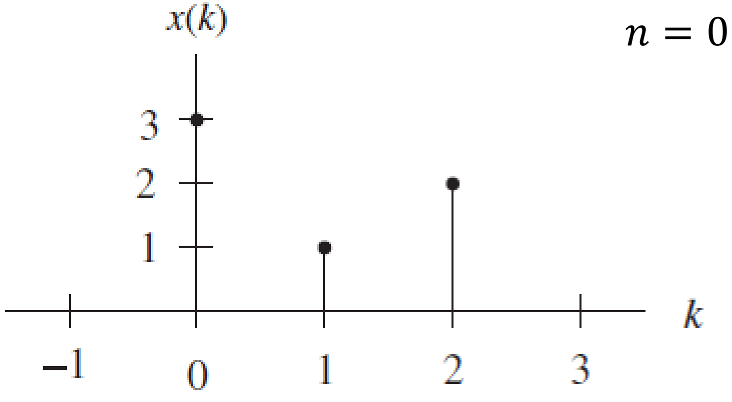
Example

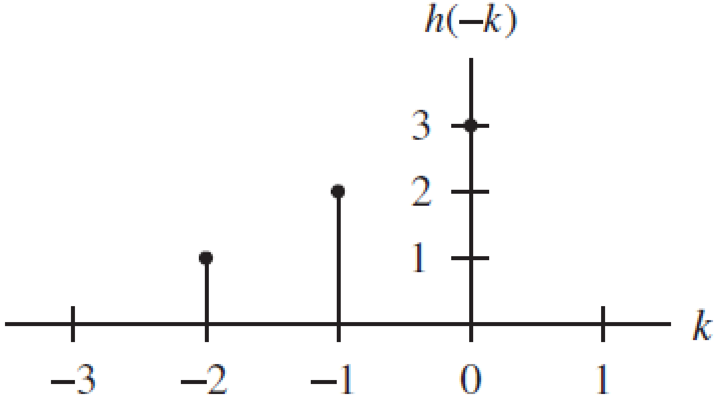
$n = 0$

The output sequence is calculated using

$$y(n) = \sum_{m=0}^{n} h(m)x(n-m)$$

For $n = 0$, we have $y(0) = h(0)x(0)$





Example

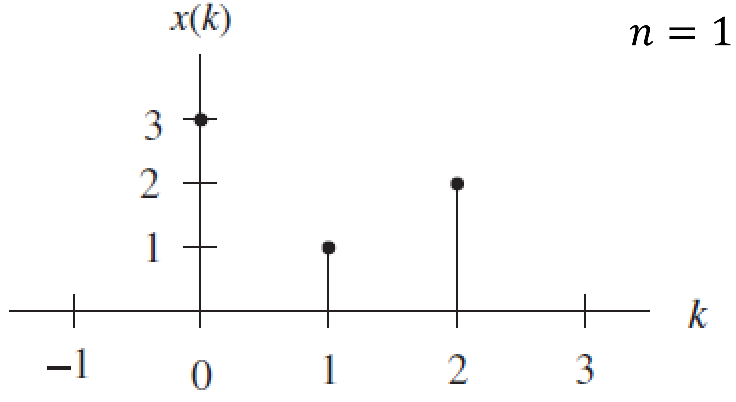
The output sequence is calculated using

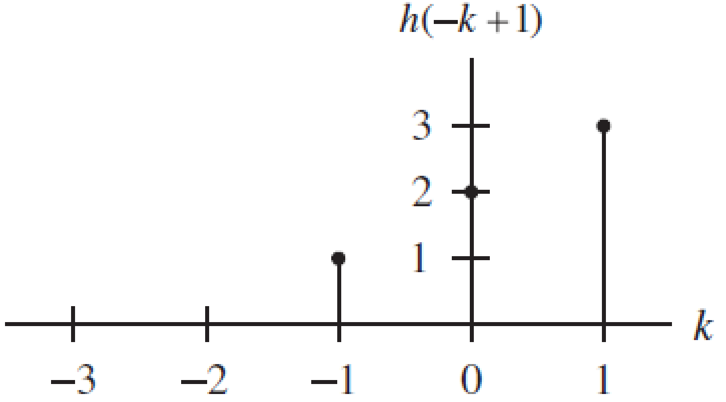
$$y(n) = \sum_{m=0}^{n} h(m)x(n-m)$$

For $n = 0$, we have $y(0) = h(0)x(0)$
For $n = 1$, we have

$$y(1) = h(0)x(1) + h(1)x(0)$$

$n = 1$





Example

The output sequence is calculated using

$$y(n) = \sum_{m=0}^{n} h(m)x(n-m)$$

For $n = 0$, we have $y(0) = h(0)x(0)$

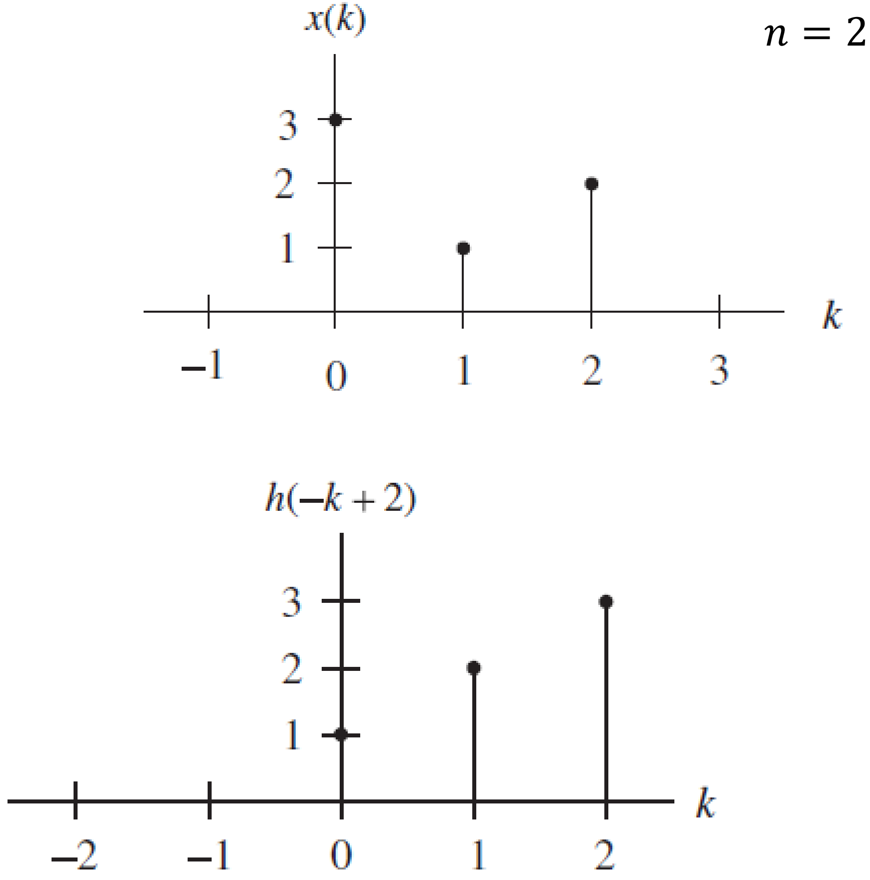
For $n = 1$, we have

$$y(1) = h(0)x(1) + h(1)x(0)$$

For $n = 2$, we have

$$y(2) = h(0)x(2) + h(1)x(1) + h(2)x(0)$$

$n = 2$



Example

The output sequence is calculated using

$$y(n) = \sum_{m=0}^{n} h(m)x(n-m)$$

For $n = 0$, we have $y(0) = h(0)x(0)$

For $n = 1$, we have

$$y(1) = h(0)x(1) + h(1)x(0)$$

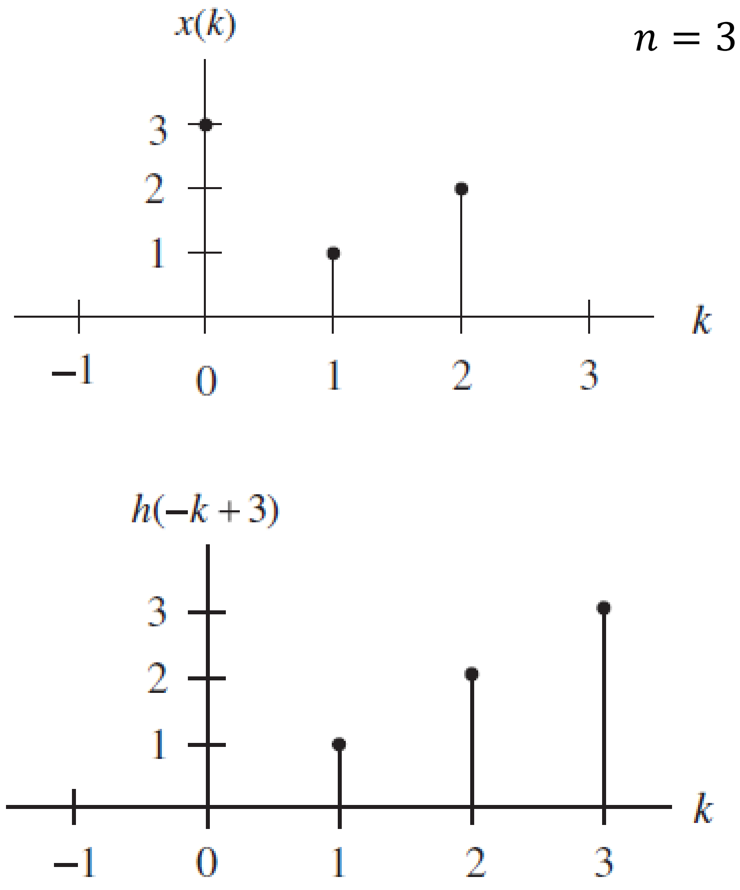
For $n = 2$, we have

$$y(2) = h(0)x(2) + h(1)x(1) + h(2)x(0)$$

For $n = 3$, we have

$$y(3) = h(0)x(3) + h(1)x(2) + h(2)x(1) + h(3)x(0)$$

$n = 3$



Example

The output sequence is calculated using

$$y(n) = \sum_{m=0}^{n} h(m)x(n-m)$$

For $n = 0$, we have $y(0) = h(0)x(0)$

For $n = 1$, we have

$$y(1) = h(0)x(1) + h(1)x(0)$$

For $n = 2$, we have

$$y(2) = h(0)x(2) + h(1)x(1) + h(2)x(0)$$

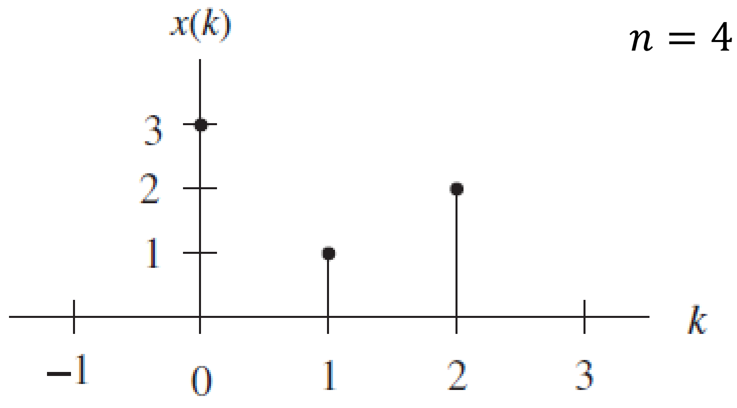
For $n = 3$, we have

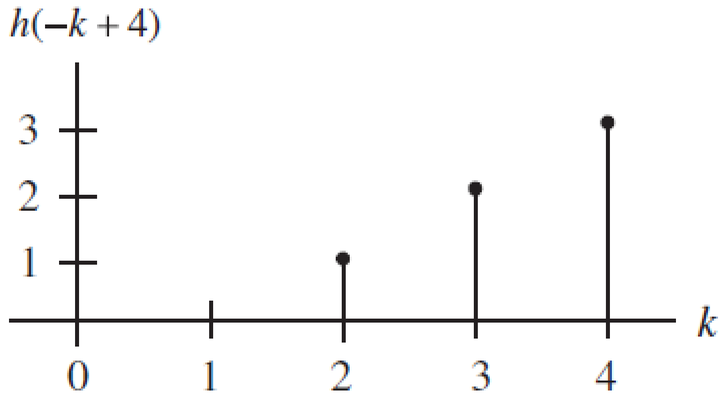
$$y(3) = h(0)x(3) + h(1)x(2) + h(2)x(1) + h(3)x(0)$$

For $n = 4$, we have

$$y(4) = h(1)x(3) + h(2)x(2) + h(3)x(1) + h(4)x(0)$$

$n = 4$





Exercise

Input signal x = [1,2,3]
Digital filter h=[1, 2]
Calculate convolution $y(n) = x(n) * h(n)$

Example & Matlab code

Input signal x = [1,2,3]
Digital filter h=[1, 2]
Calculate convolution $y(n) = x(n) * h(n)$

Answer:
$y(0) = h(0)x(0) = 1$
$y(1) = h(0)x(1) + h(1)x(0) = 4$
$y(2) = h(0)x(2) + h(1)x(1) = 7$
$y(3) = h(1)x(2) = 6$

Then you try using Matlab:

```
x = [1, 2, 3]; % Input signal
h = [1, 2]; % Filter
y = conv(x, h); % Convolution
```

Property of Convolution

→ Commutative

$$a(n) * b(n) = b(n) * a(n)$$

→ Associative

$$\big(a(n) * b(n)\big) * c(n) = a(n) * \big(b(n) * c(n)\big)$$

→ Distributive

$$a(n) * \big(b(n) + c(n)\big) = a(n) * b(n) + a(n) * c(n)$$

Basic sequence

→ Unit sample sequence (impulse)

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & oherwise \end{cases}$$

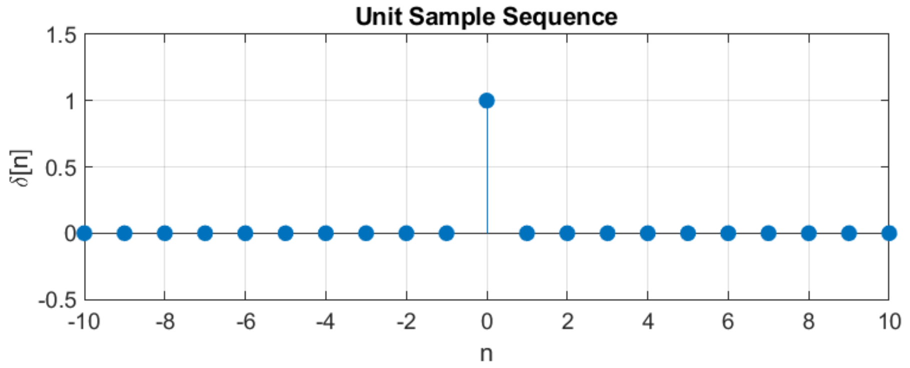
→ Sampling at time t=2, $\delta(t-2)$

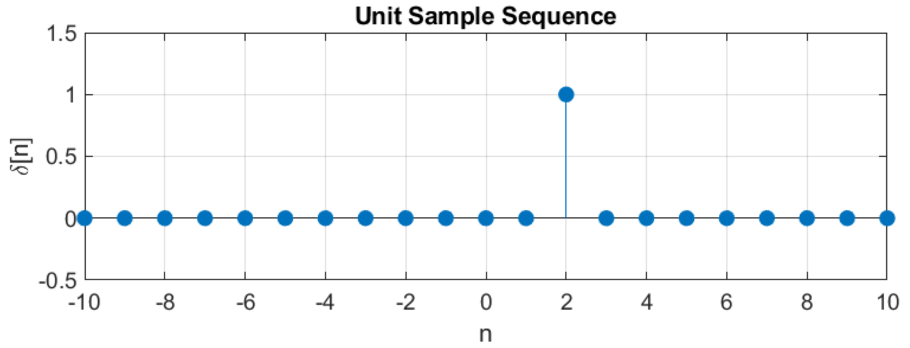
→ Shift to the right $\delta(t-1)$ -> delay

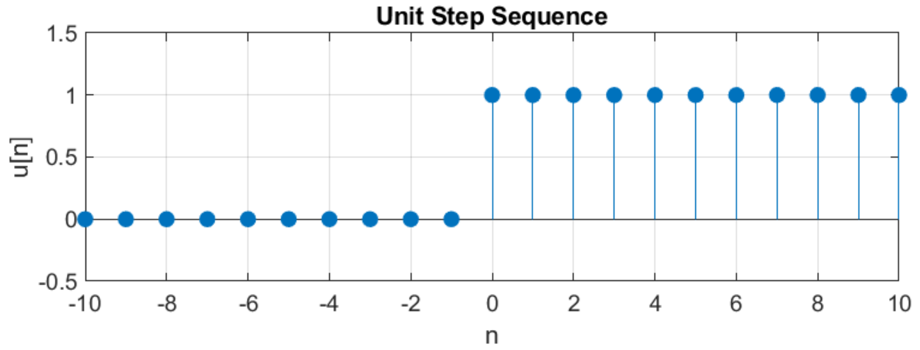
→ Shift to the left $\delta(t+1)$ -> advance

→ Step sequence

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & oherwise \end{cases}$$



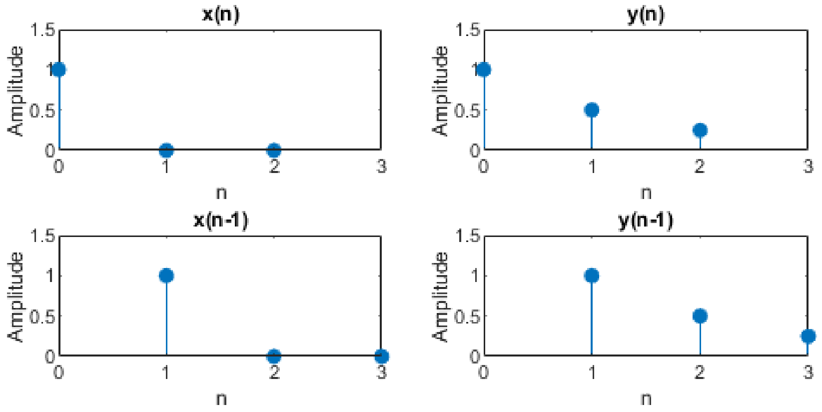




Linear Time Invariant (LTI)

Let the outputs of a system be $y_1(n)$ and $y_2(n)$ respectively, when the inputs are $x_1(n)$ and $x_2(n)$. The system is called **Linear** if the output $y(n) = ay_1(n) + by_2(n)$ in response to input $x(n) = ax_1(n) + bx_2(n)$

Let a system have output $y(n)$ when the input is $x(n)$. The system is called **time invariant** if $x(n - n_0)$ produces y$(n - n_0)$

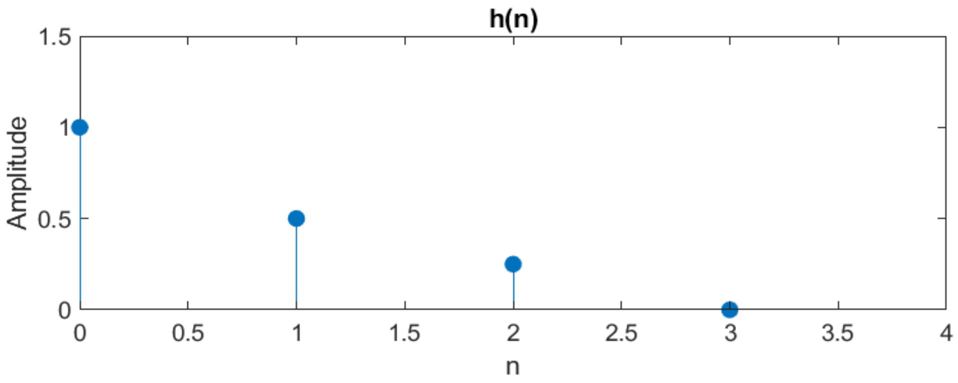


A system is **LTI (Linear and Time Invariant)** if it is both linear and time invariant

Causal

A system with input $x(n)$ and output $y(n)$ is **causal** if $y(n)$ depends only on $x(n)$, $x(n-1)$, $x(n-2)$, …, but not on $x(n+1)$, $x(n+2)$, …

The LTI system is **causal** if and only if $h(-1) = h(-2) = \cdots = 0$
i.e. $h(n) = 0,\ \forall n < 0$



SDU

Linear Constant-Coefficient Difference Equation

Consider an LTI system, whose input $x(n)$ and output $y(n)$ can be described by

$$y(n) + a_1 y(n-1) + \cdots + a_N y(n-N) = b_0 x(n) + b_1 x(n-1) + \ \ldots + b_M x(n-M)$$

$$y(n) = b_0 x(n) + b_1 x(n-1) + \ \ldots + b_M x(n-M) - a_1 y(n-1) - \cdots - a_N y(n-N)$$

$a_1, \ldots, a_N, b_0, \ldots, b_M$ are the constant coefficient

$y(n_0)$ is determined from $y(n_0-1), \ldots, y(n_0-N),\ x(n_0), \ldots, x(n_0-M)$

Example

$$y(n) = ay(n-1) + x(n)$$
$$x(n) = \delta(n)$$

In this case, $y(0) = ay(-1) + x(0)$

$$y(-1) = 0, \qquad y(n) = 0, \qquad n < 0$$

Then

$$y(0) = 1$$

And

$$y(1) = ay(0) + x(1) = a$$
$$y(2) = ay(1) + x(2) = a^2$$
$$\cdots$$

Results

$$y(n) = \begin{cases} 0, & n < 0 \\ a^n, & n \geq 0 \end{cases}$$

Example

$$y(n) = 0.25y(n-1) + x(n)$$
$$\text{for} \quad n \geq 0 \quad \text{and} \quad y(-1) = 0$$

Determine the unit-impulse response $h(n)$.

Let $x(n) = \delta(n)$,

$$h(n) = 0.25\, h(n-1) + \delta(n)$$

$h(0) = 0.25\, h(-1) + \delta(0) = 0.25 \times 0 + 1 = 1$
$h(1) = 0.25\, h(0) + \delta(1) = 0.25 \times 1 + 0 = 0.25$
$h(2) = 0.25\, h(1) + \delta(2) = 0.25 \times 0.25 + 0 = 0.0625$
$\dots$

Exercise

Book Page 80-85.

- 3.5
- 3.6
- 3.10
- 3.15
- 3.18
- 3.27